

BIS REFERENCE HANDBOOK: ELECTRICAL ENGINEERING & POWER APPARATUS

STANDARDS COMPENDIUM FOR TRANSFORMERS, CABLES, MOTORS & SWITCHGEAR

This official reference handbook provides engineers, utilities, and manufacturers with guidance on compliance with mandatory electrotechnical Indian Standards.

1. DISTRIBUTION TRANSFORMERS (IS 1180 PART 1 : 2014)

- Applicable to outdoor/indoor liquid immersed distribution transformers up to 2500 kVA, 33 kV.
- Mandatory energy efficiency loss levels: Level 1, Level 2, and Level 3 (BEE 5-star equivalent).
- Routine tests (winding resistance, voltage ratio, short-circuit impedance) and type tests (temperature rise, lightning impulse, short-circuit withstand test at CPRI/ERDA).

2. ELECTRIC CABLES & WIRES (IS 694 & IS 7098)

- **IS 694 : 2010:** PVC insulated cables for working voltages up to and including 1100 V (house wiring, flexible cords).
- **IS 7098 (Part 1 & 2):** Cross-linked Polyethylene (XLPE) insulated power cables for voltages up to 33 kV. High thermal rating (90°C continuous, 250°C short circuit).

3. THREE-PHASE INDUCTION MOTORS (IS 12615 : 2018)

Specifies energy efficiency classes for 3-phase squirrel cage induction motors (0.12 kW to 1000 kW):

Efficiency Class	Standard Code	Statutory Status in India
IE1 (Standard Efficiency)	IS 12615	Phased out in India
IE2 (High Efficiency)	IS 12615	Minimum mandatory baseline under Electrical Motors QCO
IE3 (Premium Efficiency)	IS 12615	Mandated for large industrial utilities & green procurement
IE4 (Super Premium)	IS 12615	Ultra-high efficiency for mission critical installations

4. LOW-VOLTAGE SWITCHGEAR & CONTROLGEAR (IS/IEC 60947 & IS/IEC 61439)

Covers Circuit Breakers (MCB, MCCB, ACB), Contactors, and Factory-built Low-Voltage Assemblies with internal arc containment and short-circuit withstand ratings.